

MATH

Q1) Find the circumference of a circle of radius 10.5 cm.

$$C = 2\pi r$$

$$r = 10.5 \text{ cm (given)}$$

$$\therefore \text{Circumference} = 2 \times \frac{22}{7} \times \frac{105}{10} \text{ cm}$$

$$= \boxed{66 \text{ cm}}$$

Q2) Find the circumference of a circle of radius 10 cm.
 (Take $\pi = 3.14$)

$$r = 10 \text{ cm}$$

$$C = 2\pi r$$

$$\therefore \text{Circumference} = 2 \times 3.14 \times 10 \text{ cm}$$

$$= \boxed{62.8 \text{ cm}}$$

Q) Find the diameter of a circle whose circumference is 26.4 cm

$$C = 2\pi r$$

$$\Rightarrow 26.4 = 2 \times \frac{22}{7} \times r$$

$$\Rightarrow 26.4 \times 7 = 2 \times 22 \times r$$

$$\Rightarrow r = \frac{26.4 \times 7}{2 \times 22} \text{ cm}$$

$$= \frac{42}{10} \text{ cm} = 4.2 \text{ cm}$$

another method,

$$C = 26.4 \text{ cm}$$

$$C = \pi d$$

$$\Rightarrow 26.4 = \frac{22}{7} \times d$$

$$\Rightarrow 26.4 \times 7 = 22 \times d$$

$$\Rightarrow d = \frac{26.4 \times 7}{22} = \frac{42}{10} \text{ cm}$$

$$= 4.2 \text{ cm}$$

① The circumference of a circle exceeds its diameter by 18 cm. Find the radius of a circle.

Let the radius of a circle be x cm.

ATQ,

$$C - d = 18$$

$$\Rightarrow 2\pi x - 2x = 18$$

$$\Rightarrow 2x(\pi - 1) = 18$$

$$\Rightarrow 2x\left(\frac{22}{7} - 1\right) = 18$$

$$\Rightarrow 2x\left(\frac{22-7}{7}\right) = 18$$

$$\Rightarrow 2x \times \frac{15}{7} = 18$$

$$\Rightarrow 2x \times 15 = 18 \times 7$$

$$\Rightarrow x = \frac{\cancel{18} \times 7}{\cancel{2} \times 15} = \frac{21}{5} = 4.2$$

$$\therefore x = 4.2 \text{ cm}$$

Q) The ratio the radii of the two circles is 3:4. Find the ratio of their circumferences.

Let the radii be $3u$ & $4u$ respectively and let their circumferences be C_1 & C_2 resp.

$$C_1 = 2 \times \pi \times 3u = 6\pi u \text{ \&}$$

$$C_2 = 2 \times \pi \times 4u = 8\pi u$$

Now,

$$\frac{C_1}{C_2} = \frac{6\pi u}{8\pi u} = \frac{3}{4}$$

Hence the ratio of circumference is 3:4.

Q) The diameter of a wheel of a car is 63 cm. Find the dist. travelled by the car during the period in which the wheel makes 1000 revolutions.

Diameter of the wheel = 63 cm

→ radius of the wheel = $\frac{63}{2}$ cm

→ circumference of the wheel = $2\pi r$

$$= \left(2 \times \frac{22}{7} \times \frac{63}{2} \right) \text{ cm}$$

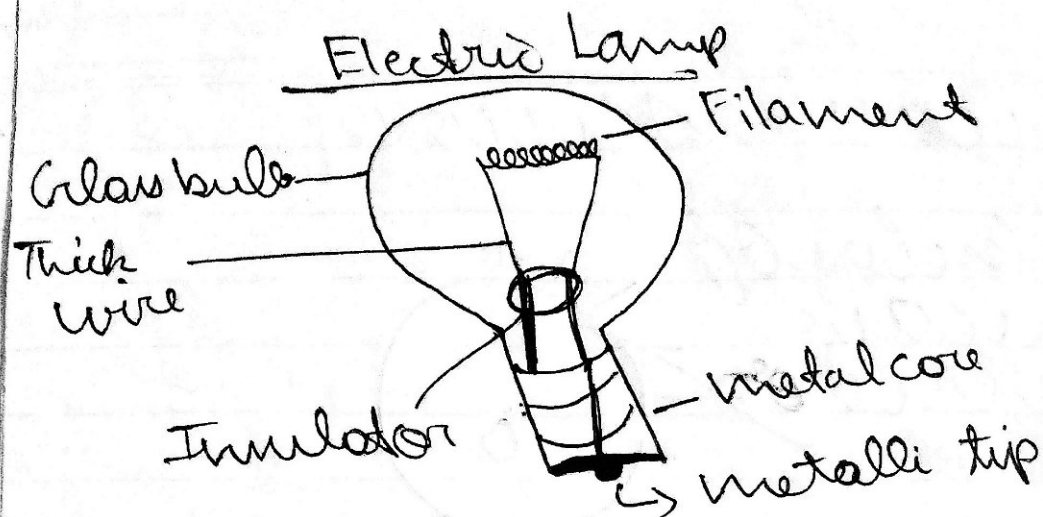
$$= 198 \text{ cm} = 1.98 \text{ m}$$

Dist. covered by the wheel in 1 revolution = 1.98 m.

Distance covered by the wheel in 1000 revolution = $\frac{198 \times 1000}{100} \text{ m}$
 $= 1980 \text{ m}$

Q) The diameter of the wheel of a car is 70 cm. How many revolutions will it make 99 km?

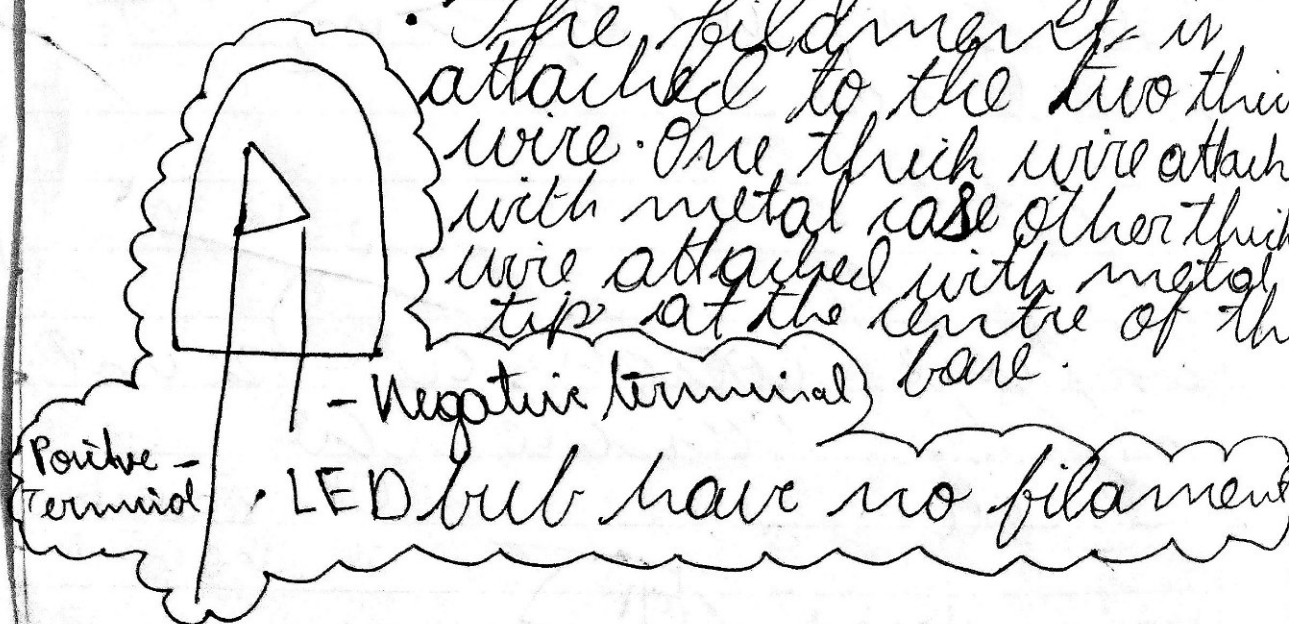
PHY.



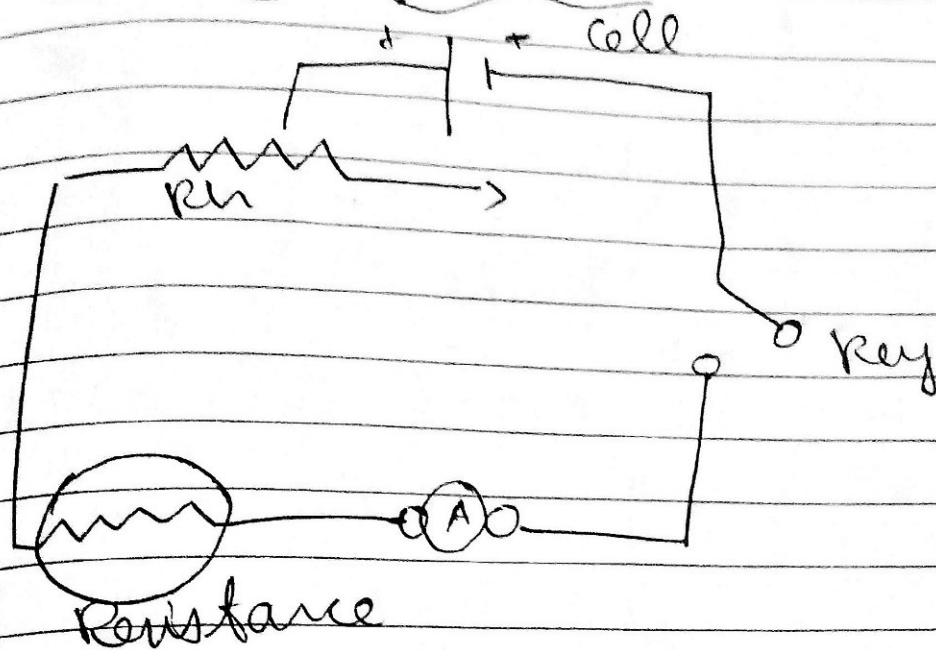
The thin wire inside the glass bulb of the lamp glows.

The glowing thin wire is called filament of the lamp.

Filament is made of tungsten.



Kind of Reflection



The positive terminal of the Ammeter is always attached the positive terminal of the cell.

AI

* Secured data transfer

Computer network uses high-speed protocols to send files from one place to another.

Components of network

1) Host - It is also known as work stations.

- It is the normal user personal computer or laptop.

- It is the basic element of a network on which user can use internet.

2) Server - It stores huge amount of data, which can be accessed by host computers.

- Server is having a high configuration hardware components.

It acts as a central computer device like a master to share information to its clients.

Server and host are connected together with wired or wireless medium.

3) Switch - It is a networking device.

It amplifies the incoming signal and distribute it to multiple ports.

It divides the whole network into smaller sub-networks.

It avoids the collision of data while transferring it from one port to another.

4) Hub - It is a networking device used to connect multiple workstations.

It is having a limitations
of 4 to 12 ports only.

Data shared from one port
using hub can be seen
at every other port.

Due to this advantage of data
sharing by hub is no longer
in use.

5) Bridge

Networking device used to
connect two similar type
of networks.

It is used to connect multiple
networking devices connected
in the same network.

It improved the data traffic
within the network.

Router

- It is used to connect two different networks together with different protocols.

- It is responsible for receiving, forwarding, rectifying.

- Router transfer the data packet based on the information saved in the routing table.

1) Gateway

- Connects two networks with different protocols.

- Acts as a entry point and exit point for a network.

- Each and every data flows through gateway in a network.

ENG.

Literary devices

- 1) Imagery → "Midnight's all a glimmer, and moon a purple glow."
- 2) Anaphora → "I will arise and go now"
- 3) Alliteration → "I hear lake water lapping with low"
- 4) Symbolism → "The pavemen grey" symbolizes the speaker's current and unfulfilling urban life.
 b) The island symbolizes a desired retreat and return to a simple and a peaceful life
- 5) Personification → "The racket rings"
 b) "peace comes dropping slow"

Metaphor → "veil of the morning" is a metaphor comparing the morning mist to veil.

b) "deep heart's core" inner peace is compared to a human heart.

Rhyming Scheme-
AB-AB

Extra Questions

1) Do you think Innisfree is only a place or a state of mind? Does the poet actually visit the place of his boyhood days?

Ans- Innisfree is a place that the poet used to visit in his boyhood. As he lives in the city now, he expresses a desire to go to Innisfree which he peaceful surroundings. This shows his state of mind.

monotony - repeating same thing

Yes, he misses the lake isle of Innisfree when he says that the sound of the lake water echoes in the core of his heart.

- Q) Briefly describe the major theme of the poem 'The Lake Isle of Innisfree'?

OR

Nature VS City Life

The contrast between nature and depressing monotony of city life is a key motif in 'The Lake Isle of Innisfree'. London serves as a metaphor for how repetitive and tiresome civilisation it is. He is not at peace because serenity only exists at Innisfree.

In contrast, Innisfree stands for nature and has a mystical aspect. One may

hear cricket's chirping.
Linnets flapping their
wings, bee buzzing over
the ripples of the lake's
water lapping against the
shore. The sky is also
amazing. In the morning
light dew drops from the
sky. At noon, the sky glows
purple, and at midnight
the stars sparkle.